

Francis Nkrumah (PhD)

Senior Lecturer & Climate Scientist

Department of Physics, University of Cape Coast, Cape Coast, Ghana

francis.nkrumah@ucc.edu.gh · +233 20 933 4438 · ORCID: 0000-0002-4723-8062

[Personal website](#) · [Institutional profile](#)

PROFILE

Climate physicist and hydro-climatologist at the University of Cape Coast with over a decade of experience analysing the West African water cycle: rainfall regimes and their trends, rainy-season onset and cessation, dry and wet spells, and extreme precipitation. Experienced in processing and analysing large geophysical datasets from rain gauges, satellites, reanalyses and climate models, and co-author of a study using a GNSS dual-frequency receiver network (TEC analysis). Hosts a GPS receiver station in the Department of Physics at UCC. Coordinator of the Meteorology and Atmospheric Physics Programme at UCC, Review Editor for the IPCC Seventh Assessment Report (Working Group I) and Contributing Author to AR6, with a strong record of multi-institutional, internationally funded research and of training African students and early-career scientists.

RESEARCH INTERESTS

- Hydro-climatology and the West African water cycle: rainfall variability, trends and seasonality
- Hydro-meteorological extremes: droughts, dry/wet spells, high-intensity rainfall, floods and coastal extreme waves
- Integrating ground-based and space-based observations (rain gauge, satellite, GNSS, reanalysis) for water-resource applications
- West African Monsoon dynamics, atmospheric moisture and mesoscale convective systems
- Regional climate modelling and climate-change impacts on water and rain-fed agriculture
- Climate intervention (SRM/SAI) impacts on regional hydro-climate

EDUCATION

2016–2020	PhD, Physics (Climate Physics) University of Cape Coast, Ghana <i>Thesis:</i> Relationship between recent rainfall regime characteristics over Southern West Africa and associated mesoscale convective systems
2012–2014	MPhil, Physics University of Cape Coast, Ghana <i>Thesis:</i> Dynamic analysis of the West African Monsoon and its variability over Ghana
2007–2011	BSc, Physics University of Cape Coast, Ghana

ACADEMIC AND PROFESSIONAL APPOINTMENTS

2023–present	Senior Lecturer & Research Scientist Department of Physics, University of Cape Coast, Ghana
2023–present	Coordinator, Meteorology and Atmospheric Physics Programme Department of Physics, University of Cape Coast, Ghana
2021–2023	Lecturer & Research Scientist Department of Physics, University of Cape Coast, Ghana
2020–2022	Postdoctoral Researcher AIMS–Canada Postdoctoral Research Fellowship in Climate Change Science, AIMS Rwanda, Kigali
2013–2021	Senior / Principal Research Assistant Department of Physics, University of Cape Coast, Ghana

RESEARCH PROJECTS AND NETWORKS

2025–present	International Lead (UCC partner team), EW4Energy — Developing capacity for storm and lightning early warning for the energy sector in Ghana
2022–present	Co-PI (Ghana team), Degrees Modelling Fund , The Degrees Initiative — regional climate impacts of stratospheric aerosol injection over West Africa
2018–present	Researcher, CORDEX-Africa (Coordinated Regional Climate Downscaling Experiment)
2015–2019	Researcher, AMMA-2050 , Future Climate for Africa (funded by UK DfID)
2012–2014	Researcher, PEER Science Project (US National Academy of Sciences / USAID)

RESEARCH INFRASTRUCTURE

Current	Host, GPS receiver station , Department of Physics, University of Cape Coast, Ghana — responsible for the on-site receiver, providing local access to GNSS observations over coastal Ghana
---------	---

INTERNATIONAL ASSESSMENTS AND PROFESSIONAL ROLES

2025–present	Review Editor , IPCC Seventh Assessment Report (AR7), Working Group I, Chapter III
2018–2024	Contributing Author , IPCC Sixth Assessment Report (AR6), Working Group I
2024–present	Member , African Climate Intervention Research Hub (ACIRH)
2017	Team Leader for Ghana , 48th International Physics Olympiad, Yogyakarta, Indonesia

SELECTED CONFERENCES AND WORKSHOPS

Jun 2026	Gordon Research Conference on Climate Engineering, Newry, Maine, USA
2026	ACIRH Workshop on Climate Intervention, Accra, Ghana
May 2025	Degrees Global Forum on Climate Intervention, Cape Town, South Africa
Apr 2023	Online Cloud Tracking Workshop, Department of Physics, University of Oxford, UK
Jan 2023	Degrees Modelling Fund Research Planning Workshop, Istanbul, Türkiye
Oct 2022	Building an African Alliance for Collaboration in Climate Science, University of Pretoria, South Africa
Jul 2022	ICTP–Africa Joint School on Rainfall and Water Resources Management, ICTP, Trieste, Italy
Jun 2019	AMMA-2050 Annual Project Meeting, Somone, Senegal (<i>presentation</i>)
Feb 2018	AMMA-2050 Annual Project Meeting, Montpellier, France (<i>presentation</i>)
Jun 2016	Eighth ICTP Workshop on the Theory and Use of Regional Climate Models, Trieste, Italy (<i>poster</i>)
May 2016	International Conference on Regional Climate (ICRC–CORDEX), Stockholm, Sweden (<i>poster</i>)
Jul 2015	Our Common Future under Climate Change, UNESCO/UPMC, Paris, France (<i>poster</i>)

PUBLICATIONS

Peer-reviewed journal articles

- [1] Quagraine, K. A., Quagraine, K. T., **Nkrumah, F.**, Tilmes, S., Hewitson, B., Tadross, M., Klutse, N. A. B. (2026). An index to characterize the South Indian Ocean high-pressure system and its variability. *Atmospheric Science Letters*. doi:10.1002/asl2.70056
- [2] Owusu, Y., Angnuureng, B. D., Kpogo-Nuwoklo, K. A., Twum, A., **Nkrumah, F.**, Nwanosike, B. (2026). Comprehensive analysis of extreme wave events along the West African coast: a study using WAVERYS data and the peak-over-threshold method. *African Geographical Review*. doi:10.1080/19376812.2025.2611302
- [3] Dommo, A., **Nkrumah, F.**, Quagraine, K. A., Klutse, N. A. B., Quenum, G. M. L. D., Koffi, H. A. (2025). Assessing the response of surface cloud radiative effects to stratospheric aerosol injections over West and Central Africa. *Journal of Geophysical Research: Atmospheres*. doi:10.1029/2025JD043576
- [4] Quagraine, K. A., Tye, M. R., Quagraine, K. T., Tilmes, S., Simpson, I. R., **Nkrumah, F.**, Egbebiyi, T. S., Odoulami, R. C., Klutse, N. A. B. (2025). Assessing the impact of stratospheric aerosol injection on precipitation extremes in Africa using the ARISE-SAI-1.5 dataset. *Environmental Research: Climate*, 4(3), 035006. doi:10.1088/2752-5295/adee3c
- [5] **Nkrumah, F.**, Quagraine, K. A., Quenum, G. M. L. D., Visioni, D., Koffi, H. A., Klutse, N. A. B. (2025). Assess-

ing regional climate trends in West Africa under geoengineering: a multimodel comparison of UKESM1 and CESM2. *Journal of Geophysical Research: Atmospheres*, 130(13), e2024JD043117. doi:10.1029/2024JD043117

- [6] **Nkrumah, F.**, Quenum, G. M. L. D., Quagraine, K. A., Tilmes, S., Klutse, N. A. B., Dommo, A., Koffi, H. A., Essien, P., Bediako, R. (2025). Climate response to stratospheric aerosol injection during the Harmattan season in West Africa. *Environmental Research: Climate*, 4(1), 015005. doi:10.1088/2752-5295/adaa0c
- [7] Quagraine, K. T., Quagraine, K. A., **Nkrumah, F.**, Egbebiyi, T. S., Klutse, N. A. B. (2025). Assessing the impact of the South Indian Ocean high-pressure system using a novel index. *International Journal of Climatology*, 45(12), e70031. doi:10.1002/joc.70031
- [8] Klutse, N. A. B., Abiodun, B. J., Quagraine, K. A., **Nkrumah, F.**, Abatan, A., Adekoke, J., Sylla, M. B., Berkoh-Oforiwa, R., Koffi, H. A., Essien, P. (2024). Projected changes in rainfall extremes over West African cities under specific global warming levels using CORDEX and NEX-GDDP datasets. *Earth Systems and Environment*. doi:10.1007/s41748-024-00425-w
- [9] Quagraine, K. A., Hewitson, B., **Nkrumah, F.**, Quagraine, K. T., Egbebiyi, T. S. (2024). A simple subtropical high-pressure system index over the South Atlantic. *Atmospheric Science Letters*. doi:10.1002/asl.1266
- [10] **Nkrumah, F.**, Klein, C., Quagraine, K. A., Berkoh-Oforiwa, R., Klutse, N. A. B., Essien, P., Quenum, G. M. L. D., Koffi, H. A. (2023). Classification of large-scale environments that drive the formation of mesoscale convective systems over southern West Africa. *Weather and Climate Dynamics*, 4(3), 773–788. doi:10.5194/wcd-4-773-2023
- [11] Essien, P., Figueiredo, C. A. O. B., Takahashi, H., Klutse, N. A. B., Wrasse, C. M., Afonso, J. M. D. S., . . . , **Nkrumah, F.**, Quagraine, K. A. (2022). Intertropical Convergence Zone as the possible source mechanism for southward propagating medium-scale traveling ionospheric disturbances over South American low-latitude and equatorial region. *Atmosphere*, 13(11), 1836. doi:10.3390/atmos13111836
GNSS dual-frequency receiver network data (2014–2019), detrended TEC maps, combined with satellite water-vapour imagery and reanalysis.
- [12] **Nkrumah, F.**, Quagraine, K. A., Quagraine, K. T., Wainwright, C., Quenum, G. M. L. D., Amankwah, A., Klutse, N. A. B. (2022). Performance of CMIP6 HighResMIP on the representation of onset and cessation of seasonal rainfall in southern West Africa. *Atmosphere*, 13(7), 999. doi:10.3390/atmos13070999
- [13] Braimah, M., Asante, V. A., Ahiataku, M. A., Ansah, S. O., Otu-Larbi, F., Yahaya, B., **Nkrumah, F.** (2022). Variability of the minor season rainfall over southern Ghana (1981–2018). *Advances in Meteorology*, 2022. doi:10.1155/2022/1861130
- [14] Klutse, N. A. B., Owusu, K., **Nkrumah, F.**, Anang, O. A. (2021). Projected rainfall changes and their implications for rainfed agriculture in northern Ghana. *Weather*. doi:10.1002/wea.4015
- [15] Quenum, G. M. L. D., **Nkrumah, F.**, Klutse, N. A. B., Sylla, M. B. (2021). Spatiotemporal changes in temperature and precipitation in West Africa. Part I: analysis with the CMIP6 historical dataset. *Water*, 13(24), 3506. doi:10.3390/w13243506
- [16] Klutse, N. A. B., Quagraine, K. A., **Nkrumah, F.**, Quagraine, K. T., Berkoh-Oforiwa, R., Dzrobi, J. F., Sylla, M. B. (2021). The climatic analysis of summer monsoon extreme precipitation events over West Africa in CMIP6 simulations. *Earth Systems and Environment*, 5(1), 25–41.
- [17] Klein, C., **Nkrumah, F.**, Taylor, C. M., Adefisan, E. A. (2021). Seasonality and trends of drivers of mesoscale convective systems in southern West Africa. *Journal of Climate*, 34(1), 71–87.
- [18] Quagraine, K. A., **Nkrumah, F.**, Klein, C., Klutse, N. A. B., Quagraine, K. T. (2020). West African summer monsoon precipitation variability as represented by reanalysis datasets. *Climate*, 8(10), 111.
- [19] Dunn, R. J., Alexander, L. V., Donat, M. G., Zhang, X., Bador, M., Herold, N., . . . (incl. **Nkrumah, F.**) (2020). Development of an updated global land in situ-based data set of temperature and precipitation extremes: HadEX3. *Journal of Geophysical Research: Atmospheres*, 125(16), e2019JD032263.
- [20] **Nkrumah, F.**, Vischel, T., Panthou, G., Klutse, N. A. B., Adukpo, D. C., Diedhiou, A. (2019). Recent trends in the daily rainfall regime in southern West Africa. *Atmosphere*, 10(12), 741.
- [21] Klutse, N. A. B., Ajayi, V., Gbobaniyi, E. O., Egbebiyi, T. S., Kouadio, K., **Nkrumah, F.**, Quagraine, K. A., Olusegun, C., Diasso, U. J., Abiodun, B. J., Lawal, K. A. A. (2018). Potential impact of 1.5 °C and 2 °C global warming on consecutive dry and wet days over West Africa. *Environmental Research Letters*.
- [22] Quagraine, K. A., Klutse, N. A. B., **Nkrumah, F.**, Adukpo, D. C., Owusu, K. (2017). Changes in rainfall characteristics in Wenchi and Saltpond farming areas of Ghana. *International Journal of Geosciences*, 8(3), 305. doi:10.4236/ijg.2017.83015

[23] **Nkrumah, F.**, Klutse, N. A. B., Adukpo, D. C., Owusu, K., Quagraine, K. A., Owusu, A., Gutowski, W. (2014). Rainfall variability over Ghana: model versus rain gauge observation. *International Journal of Geosciences*, 5, 673–683.

[24] Klutse, N. A. B., Owusu, K., Adukpo, D. C., **Nkrumah, F.**, Quagraine, K. A., Owusu, A., Gutowski, W. (2013). Farmers’ observation on climate change impacts on maize (*Zea mays*) production in a selected agroecological zone in Ghana. *Research Journal of Agriculture and Environmental Management*, 2(12), 394–402.

Preprints and assessment reports

- High-latitude, low-altitude stratospheric aerosol injection reshapes West African monsoon rainfall and associated dynamics (co-author). *Research Square* preprint (2026). [doi:10.21203/rs.3.rs-9315964/v1](https://doi.org/10.21203/rs.3.rs-9315964/v1)
- Contributing Author, IPCC (2021). *Climate Change 2021: The Physical Science Basis*. Working Group I contribution to the Sixth Assessment Report. Cambridge University Press.

PEER REVIEW

Reviewer for *Atmosphere, Remote Sensing, Water* and *Journal of Marine Science and Engineering* (MDPI).

PROFESSIONAL MEMBERSHIPS

Ghana Meteorological Society (2023–) · Ghana Science Association (2021–) · Ghana Institute of Physics (2012–) · African Climate Intervention Research Hub (2024–)

TECHNICAL SKILLS

Programming	Python, R, NCL, Ferret, Shell scripting, $\LaTeX$
Data & formats	NetCDF, GRIB, CDO, NCO; large-volume processing of gauge, satellite, reanalysis and climate-model (CMIP6, CORDEX) datasets
Methods	Trend and extreme-value analysis (incl. peak-over-threshold), rainy-season onset/cessation detection, model evaluation against observations
Systems	Linux (incl. HPC environments)

LANGUAGES

English (fluent) · Fante (fluent) · Twi (fluent)